MILITARY SPECIFICATION

# CABLE ASSEMBLIES W15-F, W16-F, W17-F and W18-F

## 1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of cable assembly entitled Cable Assemblies W15-F, W16-F, W17-F and W18-F used in test equipment for the M75 and M75E1 Adaption Kits which are used with the Nike Hercules warhead section.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

STANDARDS

MILITARY

| | |
|---|---|
| MIL-STD-105 | - Sampling Procedures and Tables for Inspection by Attributes. |
| MIL-STD-810 | - Environmental Test Methods. |
| MIL-STD-1168 | - Lot Numbering of Ammunition. |
| MIL-STD-1169 | - Packaging, Packing and Marking for Shipment of Inert Ammunition Components. |
| MIL-STD-1460 | - Soldering of Electrical Connections and Printed Wiring Assemblies, Procedures For. |

DRAWINGS

ORDNANCE CORPS

| | |
|---|---|
| EL-8817099 | - Equipment List. |
| 8822160 | - Cable Assembly, W15-F. |
| 8822161 | - Cable Assembly, W16-F. |
| 8822162 | - Cable Assembly, W18-F. |
| 8822173 | - Cable Assembly, W17-F. |
| 8822175 | - Junction Clam Shell No. 3 Breakout (Tester 41-15). |
| 8822176 | - Junction Clam Shell No. 4 Breakout (Tester 41-15). |
| 8822177 | - Junction Clam Shell No. 5 Breakout (Tester 41-15). |
| 8822178 | - Junction Clam Shell No. 6 Breakout (Tester 41-15). |



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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

## 3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The W15-F, W16-F, W17-F and W18-F cable assemblies shall comply with all requirements specified on the drawings (dwgs) 8822160, 8822161, 8822162 and 8822173, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Serialization (applicable to W15-F, W16-F and W17-F only). The contractor shall assign serial numbers as required by drawings. The contractor may select any number which complies with the drawing requirement. These numbers shall not be duplicated.

3.5 Continuity (applicable to cables W15-F, W16-F, and W17-F only). The cable assemblies shall exhibit point-to-point continuity when tested as specified in 4.4.1.

3.6 Insulation resistance (applicable to cables W15-F, W16-F and W17-F only. The insulation resistance between electrically isolated wires and between ungrounded wires and the connector shells shall not be less than 20 megohms, when a direct current (dc) potential of 500 ± 50 volts is applied when tested as specified in 4.4.2. Current to be limited to 5 ma.

3.7 Environmental testing. The cable assemblies shall comply with MIL-STD-810, visual inspection and failure criteria, section 3. The cable assemblies shall also comply with 4.3.3.1 before and after each environmental test. Testing shall be as specified in 4.4.3. Tests shall be conducted in the order specified.

3.7.1 Vibration. The cable assemblies shall withstand vibration for the frequencies, gravity unit (g) acceleration, and for the time specified in Table I, without evidence of broken or distorted parts, cracks, chipping, exposure of base material, or other damage, or impairment of operation, and shall comply with the requirements of 3.5 and 3.6 after being tested as specified in 4.4.3.1.



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Table I

| Cycle of Frequency Range (Hz) | Cycles/ Minutes | Acceleration Double Amplitude | Number of cycles Along Each Axis |
|---|---|---|---|
| 10-55 | 1 | 0.063 inch | 1 |
| 55-10 | 1 | 0.063 inch | 1 |

3.7.2 High temperature (applicable to cables W15-F, W16-F and W17-F only). The cable assemblies shall withstand a temperature of +165 + 2°F. without evidence of distortion, swelling, cracks, or broken parts and shall comply with the requirements of 3.5 and 3.6 after being tested as specified in 4.4.3.2.

3.7.3 Low temperature (applicable to cables W15-F, W16-F and W17-F only). The cable assemblies shall withstand a temperature of -65 + 2°F. without evidence of embrittlement, cracks, distortion, or broken parts and shall comply with the requirements of 3.5 and 3.6 after being tested as specified in 4.4.3.3.

3.7.4 Humidity (applicable to cables W15-F, W16-F and W17-F only). The cable assemblies shall withstand a relative humidity of 95 + 5 percent, without evidence of swelling, erosion, corrosion, pitting, cracks, exposure of base metal or other deterioration and shall comply with the requirements of 3.5 and 3.6 after being tested as specified in 4.4.3.4.

3.8 Soldering. All soldering shall comply with the requirements specified in MIL-STD-1460.

3.9 Workmanship.

3.9.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.9.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.9.3 Burrs. Parts shall be free from burrs.

3.9.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.9.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

3.9.6 Electrical. Cable assemblies and wires shall be dressed in such a manner as to prevent pinching or chafing of the wire insulation. Ends of wires shall be free from burrs, sharp edges, or points. All parts shall be free from flux, chips, dirt, grease, rust or foreign material.



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## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 3 complete assemblies (see dwgs 8822160, 8822161, 8822162 and 8822173) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table II. Tests shall be conducted in order specified in Table II.

Table II

First article inspection and tests

| Inspection or Test | Dest | Non-Dest |
|---|---|---|
| 1. Classification of characteristics (see 4.3.2.1) | | X |
| 2. Continuity (see 3.5) | | X |
| 3. Insulation resistance (see 3.6) | | X |
| 4. Vibration (see 3.7.1) | X | |
| 5. High temperature (see 3.7.2) | | X |
| 6. Low temperature (see 3.7.3) | | X |
| 7. Humidity (see 3.7.4) | | X |

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.



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4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table III and Table IV), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.



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Table III
Major Characteristics

| Lot Size | Sample Size | Lot Size | Sample Size | Lot Size | Sample Size | Lot Size | Sample Size |
|---|---|---|---|---|---|---|---|
| 1-72 | all | 140-141 | 124 | 256-259 | 177 | 541-549 | 230 |
| 73 | 72 | 142 | 125 | 260-262 | 178 | 550-559 | 231 |
| 74 | 73 | 143-144 | 126 | 263-265 | 179 | 560-569 | 232 |
| 75 | 74 | 145 | 127 | 266-268 | 180 | 570-579 | 233 |
| 76 | 75 | 146-147 | 128 | 269-271 | 181 | 580-590 | 234 |
| 77 | 76 | 148-149 | 129 | 272-275 | 182 | 591-601 | 235 |
| 78 | 77 | 150 | 130 | 276-278 | 183 | 602-613 | 236 |
| 79 | 78 | 151-152 | 131 | 279-282 | 184 | 614-625 | 237 |
| 80 | 79 | 153-154 | 132 | 283-285 | 185 | 626-637 | 238 |
| 81-82 | 80 | 155-156 | 133 | 286-289 | 186 | 638-649 | 239 |
| 83 | 81 | 157 | 134 | 290-293 | 187 | 650-662 | 240 |
| 84 | 82 | 158-159 | 135 | 294-296 | 188 | 663-676 | 241 |
| 85 | 83 | 160-161 | 136 | 297-300 | 189 | 677-690 | 242 |
| 86 | 84 | 162-163 | 137 | 301-304 | 190 | 691-704 | 243 |
| 87 | 85 | 164-165 | 138 | 305-308 | 191 | 705-719 | 244 |
| 88-89 | 86 | 166-167 | 139 | 309-312 | 192 | 720-735 | 245 |
| 90 | 87 | 168 | 140 | 313-316 | 193 | 736-751 | 246 |
| 91 | 88 | 169-170 | 141 | 317-320 | 194 | 752-768 | 247 |
| 92 | 89 | 171-172 | 142 | 321-324 | 195 | 769-785 | 248 |
| 93 | 90 | 173-174 | 143 | 325-329 | 196 | 786-804 | 249 |
| 94-95 | 91 | 175-176 | 144 | 330-333 | 197 | 805-823 | 250 |
| 96 | 92 | 177-178 | 145 | 334-338 | 198 | 824-842 | 251 |
| 97 | 93 | 179-180 | 146 | 339-342 | 199 | 843-863 | 252 |
| 98 | 94 | 181-182 | 147 | 343-347 | 200 | 864-884 | 253 |
| 99-100 | 95 | 183-184 | 148 | 348-352 | 201 | 885-907 | 254 |
| 101 | 96 | 185-187 | 149 | 353-356 | 202 | 908-930 | 255 |
| 102 | 97 | 188-189 | 150 | 357-361 | 203 | 931-955 | 256 |
| 103 | 98 | 190-191 | 151 | 362-366 | 204 | 956-981 | 257 |
| 104-105 | 99 | 192-193 | 152 | 367-372 | 205 | 982-1008 | 258 |
| 106 | 100 | 194-195 | 153 | 373-377 | 206 | 1009-1036 | 259 |
| 107 | 101 | 196-197 | 154 | 378-382 | 207 | 1037-1066 | 260 |
| 108-109 | 102 | 198-200 | 155 | 383-388 | 208 | 1067-1098 | 261 |
| 110 | 103 | 201-202 | 156 | 389-393 | 209 | 1099-1131 | 262 |
| 111 | 104 | 203-204 | 157 | 394-399 | 210 | 1132-1166 | 263 |
| 112-113 | 105 | 205-207 | 158 | 400-405 | 211 | 1167-1204 | 264 |
| 114 | 106 | 208-209 | 159 | 406-411 | 212 | 1205-1243 | 265 |
| 115 | 107 | 210-212 | 160 | 412-417 | 213 | 1244-1285 | 266 |
| 116-117 | 108 | 213-214 | 161 | 418-424 | 214 | 1286-1330 | 267 |
| 118 | 109 | 215-217 | 162 | 425-430 | 215 | 1331-1378 | 268 |
| 119 | 110 | 218-219 | 163 | 431-437 | 216 | 1379-1429 | 269 |
| 120-121 | 111 | 220-222 | 164 | 438-443 | 217 | 1430-1483 | 270 |
| 122 | 112 | 223-224 | 165 | 444-450 | 218 | 1484-1542 | 271 |
| 123-124 | 113 | 225-227 | 166 | 451-457 | 219 | 1543-1605 | 272 |
| 125 | 114 | 228-230 | 167 | 458-465 | 220 | 1606-1673 | 273 |
| 126-127 | 115 | 231-232 | 168 | 466-472 | 221 | 1674-1747 | 274 |
| 128 | 116 | 233-235 | 169 | 473-480 | 222 | 1748-1827 | 275 |
| 129-130 | 117 | 236-238 | 170 | 481-488 | 223 | 1828-1914 | 276 |
| 131 | 118 | 239-241 | 171 | 489-496 | 224 | 1915-2010 | 277 |
| 132-133 | 119 | 242-243 | 172 | 497-504 | 225 | 2011-2115 | 278 |
| 134 | 120 | 244-246 | 173 | 505-513 | 226 | 2116-2232 | 279 |
| 135-136 | 121 | 247-249 | 174 | 514-521 | 227 | 2233-2361 | 280 |
| 137 | 122 | 250-252 | 175 | 522-530 | 228 | over 2361 | 298 |
| 138-139 | 123 | 253-255 | 176 | 531-540 | 229 | | |



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Table IV
Minor Characteristics

| Lot Size | Sample Size | Lot Size | Sample Size | Lot Size | Sample Size | Lot Size | Sample Size |
|---|---|---|---|---|---|---|---|
| 1-16 | all | 28-29 | 23 | 52-56 | 31 | 132-156 | 39 |
| 17 | 16 | 30-31 | 24 | 57-62 | 32 | 157-191 | 40 |
| 18 | 17 | 32-34 | 25 | 63-69 | 33 | 192-244 | 41 |
| 19-20 | 18 | 35-36 | 26 | 70-76 | 34 | 245-334 | 42 |
| 21 | 19 | 37-40 | 27 | 77-86 | 35 | 335-519 | 43 |
| 22-23 | 20 | 41-43 | 28 | 87-97 | 36 | 520-1120 | 44 |
| 24-25 | 21 | 44-47 | 29 | 98-112 | 37 | over 1120 | 45 |
| 26-27 | 22 | 48-51 | 30 | 113-131 | 38 | | |

4.3.2.1 Classification of characteristics (see 3.2): Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Connector to cable assembly W15-F (see dwg 8822160 sheet 3 of 4 covering detail of dwg 8822160).

| Categories | Characteristics | Method of Inspection |
|---|---|---|
| Critical: | None defined. | |
| Major: | | |
| 101. | Height from top to base | Gage |
| 102. | Width of tapered body at front | Gage |
| 103. | Width of tapered body at rear | Gage |
| 104. | Height of tapered body at front | Gage |
| 105. | Height of tapered body at rear | Gage |
| 106. | Radii missing | Visual |
| *107. | Height of pin(s) (pins of the same diameter must all appear to be within 1/16 inch in height relative to the adjacent pins) | Visual/Gage |
| **108. | Depth of socket(s) (female sockets of the same diameter must all appear to be within 1/16 inch in depth relative to adjacent sockets) | Visual/Gage |
| 109. | Insulation damaged, improper or missing | Visual |
| 110. | Objective evidence relating to material missing or improper (see 3.1) | Visual |



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Minor:

201. Hardware loose, damaged or missing.....Visual/Manual
202. Any component loose, damaged or missing.....Visual/Manual
203. Any operation incomplete, incorrect or missing.....Visual
204. Evidence of poor workmanship (see 3.9).....Visual

4.3.2.1.2 Cable assembly W15-F (see dwg 8822160).

| Categories | Characteristics | Method of Inspection |
|---|---|---|

Critical: None defined.

Major:

101. Length from junction (8822176) to end of P16 cable (min).....Gage
102. Length from junction (8822176) to end of P15 cable (min).....Gage
103. Length from junction (8822176) to end of P14 cable (min).....Gage
104. Length from junction (8822176) to end of P13 cable (min).....Gage
105. Length from junction (8822177) to end of P12 cable (min).....Gage
106. Length from junction (8822177) to end of P11 cable (min).....Gage
107. Length from junction (8822177) to end of P10 cable (min).....Gage
108. Length from junction (8822177) to end of P9 cable (min).....Gage
109. Length from junction (8822177) to end of P8 cable (min).....Gage
110. Length from junction (8822178) to end of P7 cable (min).....Gage
111. Length from junction (8822178) to end of P6 cable (min).....Gage
112. Length from junction (8822178) to end of P5 cable (min).....Gage
113. Length from junction (8822178) to end of P4 cable (min).....Gage
114. Length from junction (8822178) to end of P3 cable (min).....Gage
115. Length from junction (8822178) to end of P2 cable (min).....Gage
116. Length from junction (8822175) to junction (8822177) (min).....Gage
117. Length from junction (8822175) to junction (8822178) (min).....Gage



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| | | |
|---|---|---|
| 118. | Length from junction (8822175) to junction (8822176) (min) | Gage |
| 119. | Cold soldered connections | Visual |
| 120. | Number of wire strands damaged or broken over max | Visual |
| 121. | Improper soldering (see 3.8) | Visual |
| 122. | Objective evidence relating to material missing or improper (see 3.1) | Visual |
| **Minor:** | | |
| 201. | Any component loose, damaged or missing | Visual/Manual |
| 202. | Incorrect wire in connector | Visual |
| 203. | Marking on cables unidentifiable or misleading | Visual |
| 204. | Evidence of poor workmanship (see 3.9) | Visual |
| 205. | Serialization improper or missing (see 3.4) | Visual |

4.3.2.1.3 Cable assembly W16-F (see dwg 8822161).

| Categories | Characteristics | Method of Inspection |
|---|---|---|
| Critical: | None defined. | |
| Major: | | |
| 101. | Total length (min) | Gage |
| 102. | Improper soldering (see 3.8) | Visual |
| 103. | Connector pins damaged or bent | Visual |
| 104. | Connector insulation damaged | Visual |
| *105. | Height of pin(s) (pins of the same diameter must all appear to be within 1/16 inch in height relative to the adjacent pins) | Visual/Gage |
| **106. | Depth of socket(s) (female sockets of the same diameter must all appear to be within 1/16 inch in depth relative to the adjacent sockets) | Visual/Gage |
| 107. | Objective evidence relating to material missing or improper (see 3.1) | Visual |
| Minor: | | |
| 201. | Insulation sleeving missing or damaged | Visual |
| 202. | Any component loose, damaged or missing | Visual/Manual |



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| | | |
|---|---|---|
| 203. | Marking misleading or unidentifiable | Visual |
| 204. | Evidence of poor workmanship (see 3.9) | Visual |
| 205. | Serialization improper or missing (see 3.4) | Visual |

4.3.2.1.4 Cable assembly W17-F (see dwg 8822173).

| Categories | Characteristics | Method of Inspection |
|---|---|---|
| Critical: | None defined. | |
| Major: | | |
| 101. | Improper soldering (see 3.8) | Visual |
| 102. | Connector pins damaged or bent | Visual |
| 103. | Connector insulation damaged | Visual |
| *104. | Height of pin(s) (pins of the same diameter must all appear to be within 1/16 inch in height relative to the adjacent pins) | Visual/Gage |
| **105. | Depth of socket(s) (female sockets of the same diameter must all appear to be within 1/16 inch in depth relative to the adjacent pins) | Visual/Gage |
| 106. | Total length (min) | Gage |
| 107. | Objective evidence relating to material missing or improper (see 3.1) | Visual |
| Minor: | | |
| 201. | Marking misleading or unidentifiable | Visual |
| 202. | Evidence of poor workmanship (see 3.9) | Visual |
| 203. | Any component loose, damaged or missing | Visual/Manual |
| 204. | Serialization missing or improper (see 3.4) | Visual |

4.3.2.1.5 Ground strap assembly W18-F (see dwg 8822162).

| Categories | Characteristics | Method of Inspection |
|---|---|---|
| Critical: | None defined. | |
| Major: | | |
| 101. | Improper soldering (see 3.8) | Visual |
| 102. | Excessive or loose soldering | Visual/Manual |



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103. Number of wire strands damaged or broken over max.....Visual/Manual
104. Length (min).....Gage
105. Objective evidence relating to material missing or improper (see 3.1).....Visual

Minor:
201. Marking misleading or unidentifiable...Visual
202. Evidence of poor workmanship (see 3.9).....Visual
203. Any component loose, damaged or missing.....Visual/Manual

*In the event pin(s) do not appear to be within 1/16 inch in height relative to the adjacent pins, Gage (see 4.3.4) shall be used to determine the variation from the normal height. Reject any pin that varies by more than 0.063 inch max. from the normal height of the individual connector. The normal height of the individual connector pins shall be determined by calculating the average height from measurements taken of pins adjacent to the pin that appears to be defective. The forward metal edge of the connector shall be utilized as the reference surface for all measurements of pin heights.

**In the event socket inserts do not appear to be within 1/16 inch in depth relative to the adjacent sockets, Gage (see 4.3.4) shall be used to determine the variation from the normal depth. Reject any insert that varies by more than 0.063 inch max. from the normal socket insert depth of the individual connector. The normal depth of the individual connector socket shall be determined by calculating the average depth from measurements taken of sockets adjacent to the socket that appears to be defective. The forward edge of the connector insert shall be utilized as the reference surface for all measurements of socket depths.

4.3.2.1.6 Preparation for delivery (see 5).

| Categories | Characteristics | Method of Inspection |
|---|---|---|
| Critical: | None defined. | |
| Major: | | |
| | 101. Preservation, packing and packaging not in accordance with MIL-STD-1169 | Visual |
| | 102. Marking not in accordance with MIL-STD-1169 | Visual |
| Minor: | | |
| | 201. Evidence of poor workmanship (see 3.9) | Visual |



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4.3.3 Testing. All tests shall be performed. Tests shall be performed in the order specified below:

| Test | Test Classification |
|---|---|
| a. Continuity | Major |
| b. Insulation resistance | Major |
| c. Vibration | Major |

Of these tests, (c) is considered destructive and samples so tested shall not be returned to the lot. Destructive samples are over and above the deliverable quantity of items required by the contract.

4.3.3.1 Continuity and insulation resistance. Each cable assembly shall be tested in accordance with 4.4.1 and 4.4.2. If any item fails to comply with 3.5 and 3.6, the item shall be rejected and removed from the lot.

4.3.3.2 Vibration. Two sample cable assemblies shall be selected at random from each lot. The samples shall be tested in accordance with 4.4.3.1. If any item fails to comply with 3.7.1, the lot shall be rejected.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL) (see EL-8817099). When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.



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4.4.1 Continuity. The circuit continuity between electrically connected pins and sockets shall be checked in accordance with the applicable drawing. A maximum potential of 6 volts dc shall be applied between connector contacts.

4.4.2 Insulation resistance. The required potential shall be applied and the insulation resistance measured between electrically isolated wires and between all ungrounded wires and the connector shells utilizing an insulation tester having a direct reading scale in megohms.

4.4.3 Environmental testing (see 3.7).

4.4.3.1 Vibration. The cable assemblies, with mating connectors attached, shall be mounted to a suitable vibration producing apparatus. The cables shall be firmly affixed to the vibration source at a point a maximum of six inches behind the cable connectors. The mating connectors shall be firmly attached to the fixture specified on applicable IEL. The length of the cable shall be suitably coiled so as not to interfere with the vibration apparatus. The cable assemblies, thus arranged, shall be vibrated in the two mutually perpendicular axis, perpendicular to the longitudinal axis of the connectors in accordance with the schedule specified in Table I. The frequency shall be varied uniformly throughout each frequency range specified. At the completion of the vibration test, the cable assemblies shall be tested as specified in 4.4.1 and 4.4.2.

4.4.3.2 High temperature. The cable assemblies shall be placed in a test chamber with the electrical circuits made externally available through appropriate electrical cabling. The temperature within the chamber shall be raised to +165 + 2°F. and maintained at that level until the cable has stabilized in temperature (see 6.4). Stabilization shall be determined by an appropriate thermal device in direct contact with the cable. After temperature stabilization, the cable shall be tested as specified in 4.4.2. The temperature chamber shall then be reduced to ambient room temperature. No forced cooling shall be employed, such as directing air on the cable assemblies by means of a fan or placing the cable assemblies in refrigerated air. Upon return to ambient room temperature, the cable assemblies shall be visually and manually examined and then tested as specified in 4.4.1.

4.4.3.3 Low temperature. The cable assemblies shall be tested as specified in 4.4.3.2 except that the chamber temperature shall be reduced to -65 + 2°F. before starting the test, and no forced warming, such as directing warm air on the cable by means of a fan or placing the cable in warm air, shall be used in returning the cable assemblies to ambient room temperature. After completion of the lot temperature test, the cable assemblies shall be tested as specified in 4.4.1 and 4.4.2.



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4.4.3.4 Humidity. The cable assemblies, with the dust caps in place, shall be loosely coiled and placed in a humidity test chamber. Within a period of one hour, the temperature of the test chamber shall be raised to +165 ± 2°F., stabilized (see 6.4) and maintained for a period of two hours. A relative humidity of 95 percent minimum shall be maintained while the temperature is +165 ± 2°F. The chamber temperature shall be then reduced uniformly over a period of time not exceeding one hour to between 68°F. and 95°F. During the period of decreasing temperature, the relative humidity shall be 100 percent. The cable assemblies shall be subjected to a total of 10 humidity cycles (one hour increasing temperature, two hours of temperature stabilization and 95 percent relative humidity, and one hour of decreasing temperature and 100 percent relative humidity). Distilled water, having a pH value of between 6.5 and 7.5 at 77°F., shall be used in controlling the relative humidity. The pH measurement shall be made electrically, using glass electrode with a saturated potassium chloride bridge or another method of equivalent accuracy. Only chemically pure hydrochloric acid or chemically pure sodium hydroxide shall be used to adjust the pH value. Following the tenth cycle, the cable shall be immediately removed from the chamber and excessive moisture removed by blowing dry, filtered, compressed air over the cable for a period of 5 minutes. After completion of the humidity test the cable shall be tested as specified in 4.4.1 and 4.4.3.

## 5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The cable assemblies shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169.

5.1.2 Level C. The cable assemblies shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169.

5.2 Packing.

5.2.1 Levels A and C. The cable assemblies shall be packed in accordance with the applicable instructions contained in MIL-STD-1169.

5.3 Marking. Marking shall be in accordance with MIL-STD-1169.

## 6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.
b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).



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6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-T.

6.4 Temperature stabilization. The temperature is considered stabilized when the greatest peaks (both highest and lowest) continually observed over 5 minutes min. do not exceed the nominal temperature plus tolerance specified.

6.5 Supersession data. This specification supersedes the requirements of Purchase Description X-PA-PD-1797 Rev. 1 dated 14 August 1964.

Custodian:
Army - PA

Preparing activity:
Army - PA

Project No. 1190-A071



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

**DD FORM 1 JAN 72 1426** REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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